



Physiologically Based Pharmacokinetic (PBPK) Modeling of Antidepressants: A Scoping Review of Existing Models, Applications, and Software Tools

Rebecca Leyla Huber¹ · Oliver Scherf-Clavel¹

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Abstract

Physiologically based pharmacokinetic models are a valuable tool for simulating antidepressant pharmacokinetics in patient groups that are often underrepresented in clinical trials. Clinical data on antidepressant use in vulnerable populations such as geriatric patients or pregnant women remain limited, making safe and personalized treatment challenging. The objective of this review was to systematically evaluate existing physiologically based pharmacokinetic models for antidepressants, their applications, and the modeling software platforms used, in order to identify relevant research gaps and highlight future needs for model development. A scoping review was conducted in accordance with Preferred Reporting Items for Scoping Reviews (PRISMA-ScR) guidelines. Seven electronic databases were systematically searched. Data on the modeled drug, model applications, and software platforms were extracted and evaluated. A total of 60 studies met the inclusion criteria. The availability of physiologically based pharmacokinetic models across antidepressants was heterogeneous. Paroxetine, fluvoxamine, and venlafaxine were most frequently modeled ($n = 10$ each), whereas few or no models exist for substances such as mirtazapine ($n = 1$) or milnacipran ($n = 0$). The main application areas were drug–drug interactions ($n = 18$), pregnancy ($n = 14$), and drug-(drug-)gene interactions ($n = 13$). Pediatric populations and specific ethnic groups were rarely addressed. SimCyp® Simulator was the most used modeling platform, followed by PK-Sim®/Mobi®, GastroPlus®, and R®/RStudio®. Although physiologically based pharmacokinetic models for antidepressants are increasingly available, they focus on a limited number of drugs and applications. Substantial evidence gaps remain, particularly for vulnerable patient populations.

Plain Language Summary

Antidepressants are widely used to treat conditions such as depression. However, there are limited clinical data on how these drugs work in vulnerable populations, including older adults, pregnant women, children, and people with different genetic backgrounds. Physiologically based pharmacokinetic models, or as originally stated in the manuscript, PBPK models, are computer-based tools that simulate how a drug moves through the body. These models can help predict how antidepressants behave in groups that are often underrepresented in clinical studies. This review systematically looked at existing PBPK models for antidepressants, the ways they have been used, and the software platforms employed to create them. Sixty studies were included. Most models focused on the antidepressants paroxetine, fluvoxamine, and venlafaxine, while other drugs such as mirtazapine or milnacipran had very few or no models. The main applications of these models were studying interactions with other drugs, predicting effects during pregnancy, and exploring how genetics affects drug response or drug interactions. Models for children and specific ethnic groups were rare. The most commonly used software was SimCyp® Simulator, followed by PK-Sim®/Mobi®, GastroPlus®, and R®/RStudio®. Overall, PBPK models for antidepressants are increasingly available, but they cover only a limited number of drugs and applications. Large gaps remain, especially for vulnerable patient populations, highlighting the need for future research and model development.

✉ Oliver Scherf-Clavel
O.Scherf-Clavel@lmu.de

¹ Department of Pharmacy, Ludwig-Maximilians-Universität München, Butenandtstraße 5, 81377 Munich, Germany

Key Points

Heterogeneous availability of physiologically based pharmacokinetic (PBPK) models for different antidepressants and application areas.

Most PBPK models for: paroxetine, fluvoxamine and venlafaxine.

Rare or missing PBPK models for: mirtazapine, trimipramine or milnacipran.

Scarce or no PBPK models for antidepressants addressing different ethnicities or the use of antidepressants in children.

1 Introduction

Mental disorders, including depression, anxiety disorders, and other affective or stress-related conditions, represent a major and growing global health burden with over one billion people affected worldwide. These conditions are associated with significant morbidity, reduced quality of life, and increased mortality, which highlights the need for effective and safe treatment strategies [1–3]. Pharmacotherapy plays a central role in the management of many of these disorders, with antidepressants being widely used across various psychiatric indications [4]. However, clinical studies are often limited in their ability to investigate specific scenarios, for example, because of ethical considerations, or a small number of patients meeting specific inclusion criteria, particularly in vulnerable populations or rare clinical settings. As a result, many patients treated in routine clinical practice are underrepresented in randomized controlled trials [5]. For instance, only a small number of controlled trials have been conducted in older adults, and evidence regarding the benefits and risks of antidepressant use during pregnancy and the postpartum period remains scarce because of frequent exclusion of these populations from clinical studies [6, 7].

Physiologically based pharmacokinetic (PBPK) modeling has emerged as a valuable tool in this context. As originally stated in the manuscript PBPK models provide essential insights into the use of antidepressants in special patient populations, support the development of new compounds, and facilitate the assessment of clinically relevant drug–drug interactions (DDIs), among other applications [8].

In recent years, a variety of PBPK models have been developed for different antidepressants, targeting diverse purposes and clinical questions [9–12]. Despite this growing

body of work, there is currently no systematic overview of which antidepressants have been modeled and which specific objectives have been pursued. The absence of such an overview hampers the identification of areas where PBPK modeling work already exists and, conversely, where significant research and knowledge gaps remain, such as for particular antidepressants, patient populations, or clinical scenarios.

A structured summary of the existing evidence is therefore needed to define research priorities and to map the current state of PBPK modeling for antidepressants. In this context, a scoping review provides an appropriate framework to deliver a comprehensive overview of the field.

The objective of this review is (1) to identify which antidepressants have existing PBPK models, (2) to analyze the specific objectives and applications pursued with these models, (3) to document the software platforms used in each study, and (4) to assess the model development stage as well as the model structure of the identified PBPK models. By systematically compiling this information, the review aims to map the current state of research and to highlight existing knowledge gaps that may guide future investigations.

2 Methods

2.1 Review Design and Scope

A formal protocol was not registered. However, the review methods were predefined and are described in detail in the method section.

Searches were limited to publications written in English or German and published from 1977 onwards, referring to the year from which specific PBPK terminology can be reliably found in scientific databases, as demonstrated by Lu et al. in the development of their PBPK knowledge database [13].

2.2 Eligibility Criteria

Eligible studies were required to describe a PBPK model for an antidepressant that simulated the pharmacokinetics of the drug in humans. Physiologically based pharmacokinetic models developed in animals were also considered eligible when the reported findings were relevant to the clinical use of the antidepressant in humans. Whole-body (full) PBPK models as well as minimal and semi-physiological PBPK models were included.

Regarding antidepressants, the search strategy covered all substances assigned an Anatomical Therapeutic Chemical code for antidepressants that were commercially available in Germany as of July 2025. Only original research articles were included; conference abstracts and poster abstracts were excluded. Review articles were considered only if they

contained additional information relevant to a PBPK model for an antidepressant.

Studies were eligible if they described the development of a new PBPK model, the modification or adaptation of an existing model, or the application of an existing model in a new context (e.g., integration with a pharmacodynamic model). Physiologically based pharmacokinetic models developed for antidepressants explicitly intended for indications other than depression were excluded. Furthermore, studies had to describe a PBPK model with direct or indirect clinical relevance. Integrated PBPK models of metabolites of the investigated antidepressants were not considered independent models and were therefore excluded from the evaluation.

Antidepressants used only as reference substances, methodological tools, or illustrative examples were excluded unless one of the following criteria was fulfilled:

- (1) the PBPK model yielded clinically relevant results for the antidepressant;
- (2) a new PBPK model was developed, or an existing model was significantly adapted or modified;
- (3) the pharmacokinetics of the antidepressant represented the primary objective in the study.

Physiologically based pharmacokinetic model libraries provided by platforms, such as SimCyp® PBPK Simulator (Certara UK Limited, Simcyp Division, Sheffield, UK) or PK-Sim® (Open Systems Pharmacology, Leverkusen, Germany), were not systematically searched for antidepressants, as entries typically consist solely of compound files rather than application-specific models.

2.3 Search and Screening

The literature search was conducted across seven electronic databases: PubMed, Embase (via Ovid), Cochrane Library, Web of Science, PubPharm, ClinicalTrials.gov, and the Open Systems Pharmacology-based-publications-and-content (via GitHub). The specific search strategies applied in each database are provided in the Electronic Supplementary Material (ESM).

During the screening process, eight additional studies were identified through citations in previously included articles or through publications after the literature search. In accordance with the Preferred Reporting Items for Scoping Reviews (PRISMA-ScR) guidelines, no systematic backward citation search was undertaken. The review instead relied on a comprehensive, predefined database search strategy, which was considered sufficient to capture the full spectrum of existing PBPK models for antidepressants.

The screening process was conducted following the PRISMA-ScR guidelines and was carried out using the review management software Rayyan® (Rayyan Systems

Inc.; <https://www.rayyan.ai/>). The results from all database searches were imported into Rayyan® for processing.

In the first step, duplicates were automatically identified, subsequently verified, and removed manually by the authors. After deduplication, titles were screened for relevance, followed by abstract screening of the remaining records. The remaining articles were then sought for retrieval. Finally, all retrieved articles were assessed for eligibility according to the predefined inclusion and exclusion criteria. Title and abstract screening was mainly carried out by a single reviewer. For quality assurance and reducing selection bias, a second independent reviewer screened in addition 10% of all titles and abstracts.

2.4 Data Extraction and Analysis

For each included study, relevant data were extracted and recorded in a pre-developed Excel spreadsheet (Microsoft Excel, Version 2509, Microsoft 365 Apps for Enterprise; Microsoft Corporation, Redmond, WA, USA). Data extraction was primarily performed by a single reviewer, with the second reviewer independently extracting and verifying at least 10% of the data. In cases of uncertainty regarding data that had not been independently verified, the second reviewer was consulted to ensure consistency. As all relevant information could be obtained directly from the articles, no contact with the study authors was necessary. Relevant data extracted were: first author, year of publication, antidepressant(s) for which a PBPK model was developed, purpose or application of the PBPK model, the modeling software used by the authors, the model type and development stage, and whether active metabolites were included in the modeling process.

In accordance with the PRISMA-ScR guidelines, no formal assessment of the methodological quality of the included studies was performed. This scoping review was primarily intended to identify research gaps and provide a comprehensive overview of the currently available PBPK models for antidepressants. However, a critical assessment of study quality may be relevant for a future systematic review aiming to evaluate and compare the reliability of the PBPK models developed in these studies.

The extracted data were summarized in standardized tables and organized into predefined categories. Descriptive statistics were calculated for the number of PBPK models per antidepressant, and the number of PBPK models per purpose or application. These results were subsequently presented graphically.

Model development stages were assessed by first developing a score based on predefined criteria and subsequently applying it to the published models. Overall, the score comprises three weighted domains: (1) Development/Refinement/Parameter Origin (0–6 points), distinguishing between

bottom-up, middle-out, and top-down modeling approaches; (2) Robustness/Sensitivity (0–3 points), capturing the presence and type of sensitivity analysis (local, global, or none); and (3) Validation (0–11 points), which is further subdivided into type of validation (external, internal, or none; 0–5 points), scope of validation (number of clinical studies in human, dose levels, formulations, and routes of administration covered within the validation process; 0–4 points), and validation of the final model application (0–2 points). Except for the final subcategory, all components of the score refer to the entire model development lifecycle, including base models, intermediate stages, and final application model. The scoring system ranged from 0 to a maximum of 20 points. Based on the total score, model development stage was classified into five categories: very extensive (17–20 points), extensive (13–16 points), substantial (9–12 points), moderate (5–8 points), and initial (0–4 points).

Narrative summaries were employed to provide additional information on the frequency of antidepressants being investigated as perpetrator or victim drugs in DDI studies when they were not the primary drug of interest, and to give an overview of existing review articles on the topic. Figures and tables were created using Microsoft Excel (Microsoft Excel, Version 2509, Microsoft 365 Apps for Enterprise; Microsoft Corporation) to facilitate a clear and comprehensive presentation of the results.

The large language model ChatGPT (GPT-5-mini, OpenAI, 2026) and the AI-assisted writing tool Grammarly (Grammarly, Inc., 2026) were used to improve the clarity, grammar, and coherence of paragraphs throughout the paper. Data extraction for the assessment of the model development stage was supported by the use of the large language model Claude (Claude Sonnet 4.6, Anthropic) within a structured human-verified workflow. A detailed description of the extraction procedure, including prompt development, quality control, and validation steps, is provided in the ESM.

3 Results

3.1 Study Selection

The literature search yielded 2823 records, of which 867 were duplicates. Following title screening, 1327 records were excluded. Abstract screening led to the exclusion of an additional 265 articles. The remaining 364 articles were sought for retrieval; 55 articles had no full text available. All retrieved articles were then assessed for eligibility.

A total of 253 articles were excluded for the following reasons: no antidepressant included (wrong drug) [$n = 58$], no (relevant) PBPK model ($n = 43$), antidepressant not the drug of interest ($n = 62$), PBPK model explicitly developed for indications other than depression ($n = 3$), antidepressant

used solely as a methodological tool, test compound, or reference substance ($n = 71$), and review articles or other publications providing no additional information on PBPK models ($n = 16$).

During the screening process, eight additional studies were identified either through non-systematic backward citation searching of included studies or by monitoring newly published studies via databases and professional networks. Of these, four met the eligibility criteria and were subsequently included, while the remaining four were excluded because they did not investigate an antidepressant as the drug of interest.

Ultimately, 60 studies were included in the review. The overall study selection process is summarized in Fig. 1.

3.2 Data Extraction

The following variables were systematically extracted from each included study: first author, year of publication, antidepressant(s) investigated, intended purposes and applications of the PBPK model, the software platform used for model development, the model development stage as well as the structure of the respective PBPK model and whether active metabolites were modeled or not. An overview of the extracted data is presented in Table 1.

The identified PBPK models were unevenly distributed across antidepressants. Paroxetine, fluvoxamine, and venlafaxine were the most extensively modeled compounds, each with ten PBPK models. In contrast, for several antidepressants, including mirtazapine, trimipramine, and milnacipran, only a single PBPK model or no PBPK models at all were identified (see also Fig. 2).

As described in the methods section, previously developed PBPK models were also included, if they were applied in a new context. These models are specifically indicated in Fig. 2 to allow a clearer distinction between newly developed and pre-existing models and to provide a more comprehensive overview of the available modeling approaches in the field of antidepressants.

3.3 Purposes and Applications of PBPK Models

Analysis of the identified PBPK models, based on their intended purpose and application, revealed clear patterns. Physiologically based pharmacokinetic models were most frequently developed for simulating DDIs (18 models), pregnancy (14 models), and drug-(drug-)gene interactions (13 models). In contrast, only two PBPK models addressed specific ethnic groups. No PBPK model was identified for the use of antidepressants in pediatric patients for the treatment of depression (see Fig. 3).

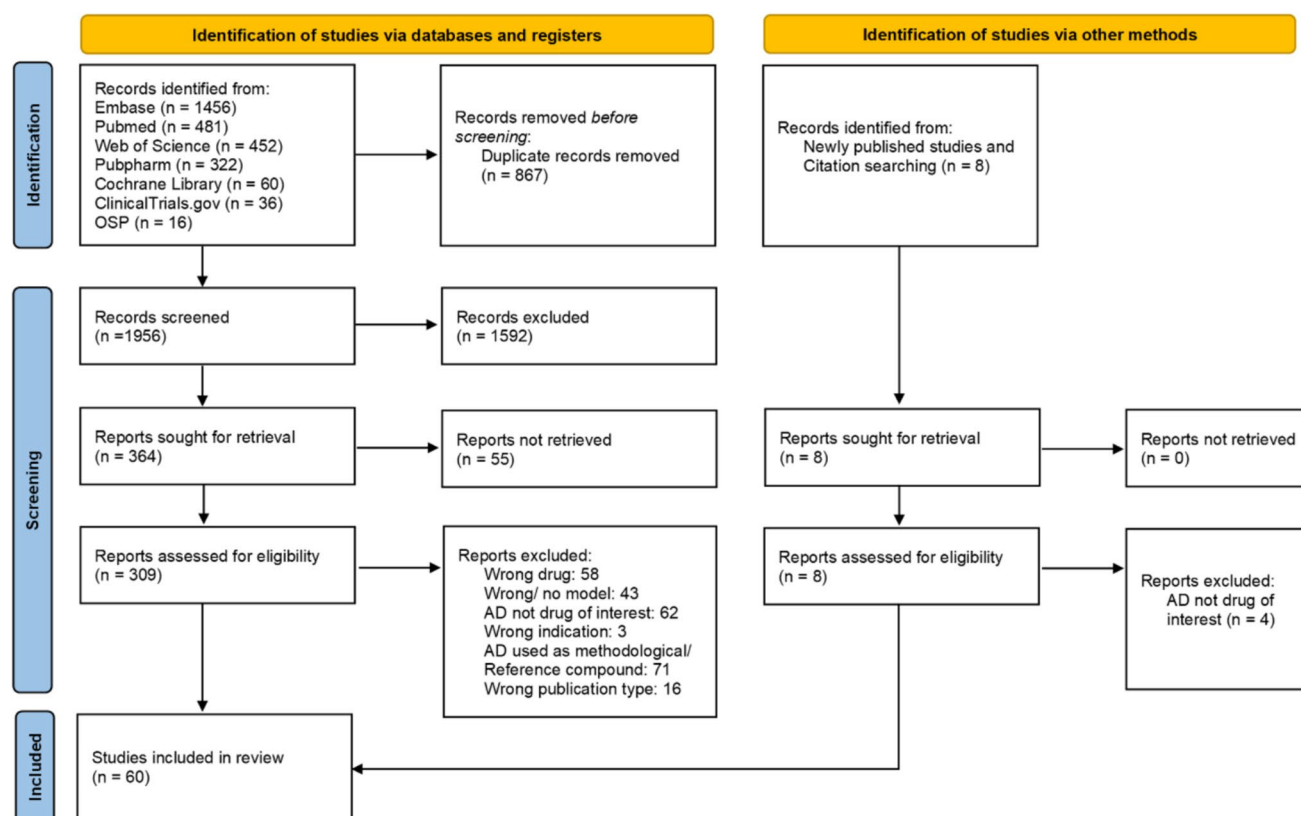


Fig. 1 Preferred Reporting Items for Systematic reviews and Meta-Analyses extension for scoping reviews (PRISMA-ScR) flow diagram. AD antidepressant, OSP Open Systems Pharmacology

Some studies addressed more than one modeling purpose. In these cases, the primary modeling purpose was used for the analysis.

It should be noted that the number of studies addressing a specific modeling purpose may differ substantially from the number of PBPK models identified for that purpose. This discrepancy occurs because some studies developed multiple PBPK models within the same modeling context. A notable example is the study conducted by Wu et al. (2025), which developed PBPK models for a total of 50 drugs intended for use in an elderly (Chinese) population, including ten antidepressants.

An analysis of model purposes and applications stratified by individual antidepressants provided more profound insights into both the current state of research and existing evidence gaps. For amitriptyline, four of the seven identified PBPK models focused on cardiac pharmacokinetics. A closer examination of these models revealed that they originated from a cardiac PBPK framework developed by Tylutki et al. [15–18], which has been successively extended and refined over time. Although these models are based on the same framework, the adaptations and extensions of the models differ in terms of complexity, model characteristics, and the context in which the models are used. Consequently, they

were considered distinct models and included separately in this review.

Despite the depth of modeling within specific application areas, this detailed assessment highlights substantial gaps in the broader research landscape, as PBPK models for many antidepressants tend to concentrate on only a limited number of clinical questions. A comparable pattern was identified for paroxetine and sertraline: the majority of PBPK models for sertraline focused on the pharmacokinetics of antidepressants during pregnancy, whereas models for paroxetine primarily referred to pregnancy and drug-(drug)-gene interactions. In contrast, other clinically relevant contexts, such as breastfeeding (lactation) or geriatric populations, were represented by only a single model each. An overview of the specific model objectives and application areas for currently available PBPK models, stratified by antidepressant, is presented in Table 2.

3.4 Software Platforms

Simcyp® Simulator (Certara, Sheffield, UK), PK-Sim®/Mobi® (Open Systems Pharmacology Suite, Leverkusen, Germany), and GastroPlus® (Simulations Plus, Lancaster, CA, USA) (among others) are dedicated PBPK platforms

Table 1 Extracted data of eligible studies. PBPK models were arranged by drug (in alphabetical order) and, within each drug, chronologically from the most recent to the oldest

Antidepressant	First author	Year	Model purpose/application	Software	Model type	Model development stage	Active metabolite(s) modeled
Amitriptyline	Scherf-Clavel [14]	2025	Pregnancy	PK-Sim®	Whole body	Extensive	Yes
	Wu [9]	2025	Geriatric	GastroPlus®	Whole body	Intermediate	No
	Tylutki [15]	2018a	Cardiac	R®	Whole body	Substantial	Yes
	Tylutki [16]	2018b	Cardiac	R®	Whole body	Extensive	Yes
	Tylutki [17]	2017	Cardiac	R®	Whole body	Intermediate	No
	Tylutki ^a [18]	2016	Cardiac	SimCyp®	n.a.	n.a.	Yes
	Howell [19]	2010	Toxicity/drug overdose	MATLAB®	Whole body	Substantial	No
Bupropion	Luecht [20]	2025	DDI (inhibitor)	PK-Sim®/MoBi®	Whole body	Intermediate	Yes
	Rüdesheim [21]	2025	D(D)GI (inhibitor)	PK-Sim®/MoBi®	Whole body	Extensive	Yes
	Zhang [22]	2022	Lactation	SimCyp®	Whole body	Intermediate	No
	Hsieh [23]	2021	Drug formulation	R®	Semi-physiological PBPK	Substantial	No
	Marok [24]	2021	D(D)GI (victim)	PK-Sim®/MoBi®	Whole body	Extensive	Yes
	Xue [25]	2017	DDI (inhibitor)	SimCyp®	Whole body	Substantial	Yes
	Barter [26]	2013	ethnicity	SimCyp®	Minimal PBPK	Substantial	Yes
Citalopram	Wu [9]	2025	Geriatric	GastroPlus®	Whole body	Intermediate	No
	Wu [27]	2020	Geriatric (+ pharmacogenetics)	SimCyp®	Whole body	Extensive	No
	Siccardi [12]	2013	DDI (victim)	SimCyp®	Not clearly determinable	Intermediate	No
Clomipramine	Uchaipichat [28]	2023	DDI (inhibitor)	SimCyp®	Minimal PBPK	Substantial	No
Desvenlafaxine	Carapeto [29]	2023	Drug formulation	GastroPlus®	Semi-physiological PBPK	Substantial	No
	Franek [30]	2015	Drug formulation	SimCyp®	Semi-physiological PBPK	Extensive	No
Doxepin	Wu [9]	2025	Geriatric	GastroPlus®	Whole body	Intermediate	No
Duloxetine	Adachi [31]	2021	toxicity/drug overdose	None	Minimal PBPK	Intermediate	No
	Storelli [11]	2019	D(D)GI (inhibitor)	SimCyp®	Whole body	Extensive	No
	Kielbasa [32]	2012	brain exposure/ CNS targeting	NONMEM®	Semi-physiological PBPK	Initial	No
Escitalopram	Choi [33]	2025	Pregnancy (+ pharmacogenetics)	SimCyp®	Whole body	Extensive	No
	Jang [34]	2025	Geriatric (+ pharmacogenetics)	SimCyp®	Whole body	Substantial	No
	Kayesh [35]	2025	Lactation	SimCyp®	Minimal PBPK	Substantial	No
	Préta [10]	2025	Pregnancy	SimCyp®	Whole body	Extensive	No
	Cardoso ^a [36]	2023	Lactation	PK-Sim®	Whole body	Substantial	No
	Delaney [37]	2018	Lactation	PK-Sim®	Whole body	Substantial	No
Esketamine	Willemin [38]	2022	DDI (inductor, inhibitor, victim)	SimCyp®	Whole body	Extensive	Yes
Fluoxetine	Spaenig [39]	2025	Toxicity/drug overdose	R®	Whole body	Extensive	No
	Wu [9]	2025	Geriatric	GastroPlus®	Whole body	Intermediate	No
	Pansari ^a [40]	2022	Lactation	SimCyp®	Minimal PBPK	Extensive	No
	Jeong [41]	2021	Ethnicity	SimCyp®	Whole body	Intermediate	yes
	Storelli [11]	2019	D(D)GI (inhibitor)	SimCyp®	Minimal PBPK	Substantial	No
	Sager [42]	2014	DDI (inhibitor)	SimCyp®	Semi-physiological PBPK	Substantial	Yes

Table 1 (continued)

Antidepressant	First author	Year	Model purpose/appli- cation	Software	Model type	Model development stage	Active metabolite(s) modeled
Fluvoxamine	Siccardi [12]	2013	DDI (victim)	SimCyp®	Not clearly determi- nable	Intermediate	No
	Rüdesheim [21]	2025	D(D)GI (inhibitor)	PK-Sim®/MoBi®	Whole body	Extensive	No
	Wu [9]	2025	Geriatric	GastroPlus®	Whole body	Intermediate	No
	Marques ^a [43]	2023	DDI (inhibitor)	GastroPlus®	Not clearly determi- nable	n.a.	No
	Burhanuddin [44]	2022	Pregnancy (+ pharma- cogenetics)	SimCyp®	Whole body	Substantial	No
	Kanacher [45]	2020	D(D)GI (inhibitor)	PK-Sim®	Whole body	Extensive	No
	Britz [46]	2019	DDI (inhibitor)	PK-Sim®/MoBi®	Whole body	Extensive	No
	Matsuoka [47]	2017	Pregnancy	MLAB®	Semi-physiological PBPK	Intermediate	No
Imipramine	Alqahtani [48]	2016	Brain exposure/ CNS targeting (+ pharma- cogenetics)	GastroPlus®	Whole body	Substantial	No
	Iga [49]	2014	DDI (inhibitor)	None	Minimal PBPK	Substantial	No
	Geldof [50]	2007	Brain exposure/ CNS targeting	None	Semi-physiological PBPK	Intermediate	No
	Pardridge [51]	2023	Brain exposure/ CNS targeting	Mathematica®	Semi-physiological PBPK	Intermediate	No
	Uchaipichat ^a [28]	2023	DDI (inhibitor)	SimCyp®	Not clearly determi- nable	n.a.	No
Mirtazapine	Iga [49]	2014	DDI (victim)	None	Minimal PBPK	Substantial	No
	Siccardi [12]	2013	DDI (victim)	SimCyp®	Not clearly determi- nable	Intermediate	No
Moclobemide	Wu [9]	2025	Geriatric	GastroPlus®	Whole body	Intermediate	No
	Kanacher [45]	2020	D(D)GI (inhibitor + victim)	PK-Sim®	Whole body	Extensive	No
Nortriptyline	Wu [9]	2025	Geriatric	GastroPlus®	Whole body	Intermediate	No
	Jornil [52]	2011	Toxicity/drug over- dose (+ pharmaco- genetics)	SimCyp®	Minimal PBPK	Substantial	No
Paroxetine	Préta ^a [10]	2025	Pregnancy	SimCyp®	Whole body	Extensive	No
	Wu [9]	2025	Geriatric	GastroPlus®	Whole body	Intermediate	No
	Pansari ^a [40]	2022	Lactation	SimCyp®	Minimal PBPK	Extensive	No
	Rüdesheim [53]	2022	D(D)GI	PK-Sim®	Whole body	Extensive	No
	Abduljalil ^a [54]	2020	Pregnancy	SimCyp®	Whole body	Extensive	No
	Almurjan [55]	2020	Pregnancy (+ pharma- cogenetics)	SimCyp®	Whole body	Substantial	No
	Storelli [11]	2019	D(D)GI (inhibitor)	SimCyp®	Minimal PBPK	Substantial	No
	Nagai [56]	2013	Pregnancy	None	Semi-physiological PBPK (in a broader sense)	Substantial	No
	Yeo [57]	2013	D(D)GI (inhibitor)	SimCyp®	Whole body	Substantial	No
	Jornil [58]	2009	D(D)GI	SimCyp®	Not clearly determi- nable	Substantial	No
Sertraline	Préta [10]	2025	Pregnancy	SimCyp®	Whole body	Extensive	No
	Wu [9]	2025	Geriatric	GastroPlus®	Whole body	Intermediate	No
	Lartey [59]	2023	Pregnancy	R®	Minimal PBPK	Extensive	No
	Nauwelaerts [60]	2023	Lactation	PK-Sim®/MoBi®	Whole body	Extensive	No
	Almurjan [61]	2021	Pregnancy (+ pharma- cogenetics)	SimCyp®	Whole body	Extensive	No

Table 1 (continued)

Antidepressant	First author	Year	Model purpose/appli- cation	Software	Model type	Model development stage	Active metabolite(s) modeled
	George [62]	2020	Pregnancy	R®	Minimal PBPK	Substantial	No
	Templeton [63]	2016	DDI (inhibitor)	n.a.	Whole body	Substantial	Yes
	Siccardi [12]	2013	DDI (victim)	SimCyp®	Not clearly determi- nable	Intermediate	No
Trazodone	Oggianu [64]	2022	Brain exposure/ CNS targeting	SimCyp®	Whole body	Substantial	No
Venlafaxine	Adachi [31]	2021	toxicity/drug overdose	None	Minimal PBPK	Intermediate	No
	Luecht [20]	2025	DDI (victim)	PK-Sim®/MoBi®	Whole body	Substantial	Yes
	Pan [65]	2025	Lactation (+ pharma- cogenetics)	SimCyp®	Whole body	Extensive	Yes
	Wu [9]	2025	Geriatric	GastroPlus®	Whole body	Intermediate	No
	Cho [66]	2024	D(D)GI (victim)	PK-Sim®	Whole body	extensive	yes
	Shen [67]	2024	D(D)GI	PK-Sim®/MoBi®	Whole body	Extensive	Yes
	Alenezi [68]	2023	Pregnancy (+ pharma- cogenetics)	SimCyp®	Whole body	Extensive	No
	Pansari [40]	2022	Lactation	SimCyp®	Minimal PBPK	Extensive	No
	Xue [25]	2017	DDI (victim)	SimCyp®	Minimal PBPK	Substantial	No
	Lin [69]	2016	Drug formulation	GastroPlus®	Semi-physiological PBPK	Substantial	Yes
	Siccardi [12]	2013	DDI (victim)	SimCyp®	Not clearly determi- nable	Intermediate	No

CNS central nervous system, DDI drug–drug interactions, D(D)GI drug-(drug-)gene interaction, n.a. not applicable, PBPK physiologically based pharmacokinetic

^aPre-existing PBPK model applied in a new context

Fig. 2 Number of physiologi-
cally based pharmacokinetic
(PBPK) models identified per
antidepressant

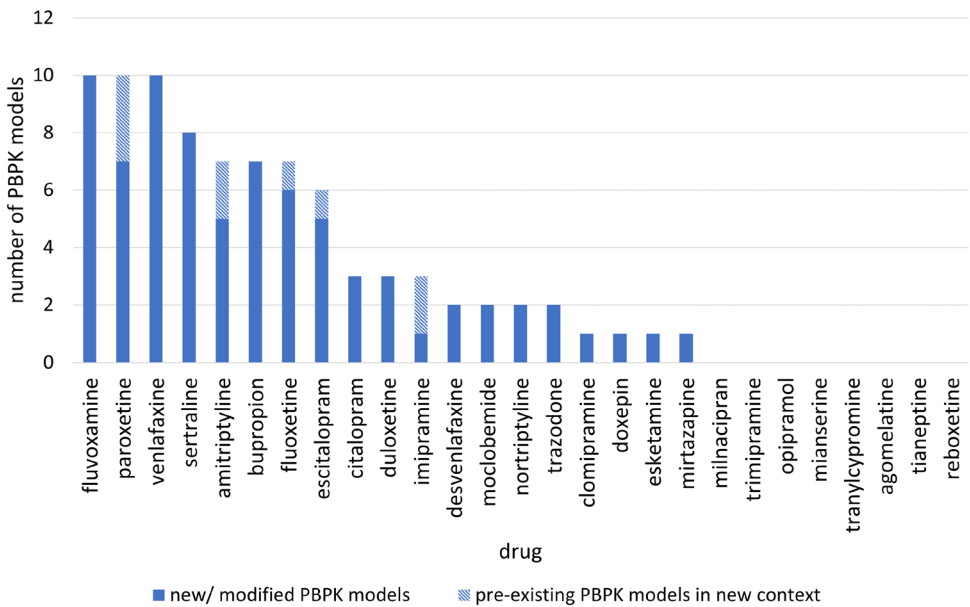


Fig. 3 Number of physiologically based pharmacokinetic (PBPK) models per model purpose/application. *CNS* central nervous system, *DDI* drug–drug interaction, *D(D)GI* drug-(drug-)gene interaction

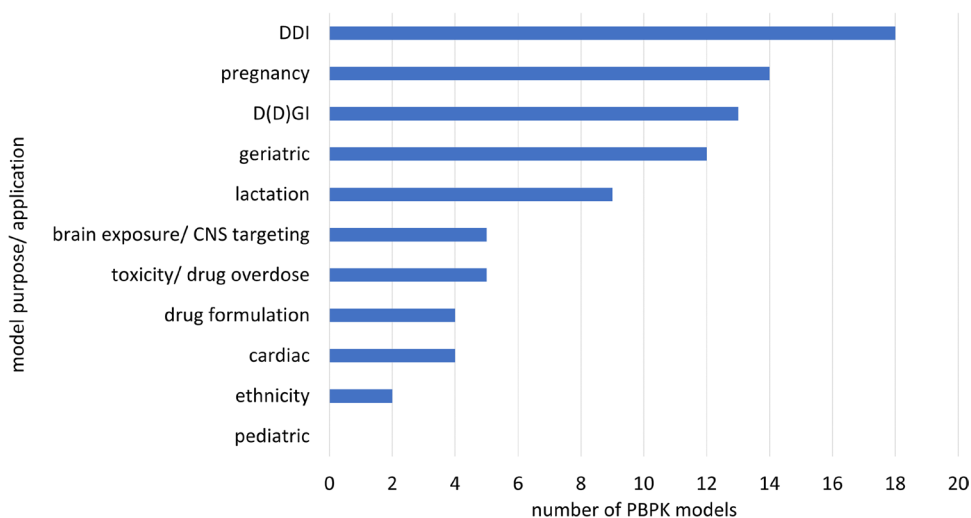


Table 2 Number of PBPK models per drug and purpose/application

	Pregnancy	DDI	D(D)GI	Lactation	Drug formulation	CNS	Cardiac	Toxicity/drug overdose	Ethnicity	Geriatric
Fluvoxamine	2	3	2			2				1
Paroxetine	4		4	1						1
Venlafaxine	1	3	2	2	1					1
Sertraline	4	2		1						1
Amitriptyline	1						4	1		1
Bupropion		2	2	1	1				1	
Fluoxetine		2	1	1				1	1	1
Escitalopram	2			3						1
Citalopram		1								2
Duloxetine			1			1		1		
Desvenlafaxine					2					
Imipramine		2				1				
Moclobemide			1							1
Nortriptyline								1		1
Trazodone						1		1		
Clomipramine		1								
Doxepin										1
Esketamine		1								
Mirtazapine		1								

CNS central nervous system, *DDI* drug–drug interaction, *D(D)GI* drug-(drug-)gene interaction, *PBPK* physiologically based pharmacokinetic model

Empty cells indicate that there was no PBPK model identified for this specific application

that provide a preconfigured modeling environment, including predefined physiological structures, compound parameter libraries, and simulation workflows. As such, they allow users to build and run PBPK models within a largely standardized, GUI-based framework. General-purpose mathematical programming environments such as R® (R Foundation for Statistical Computing, Vienna, Austria), in contrast, offer a more flexible but less preconfigured infrastructure,

providing numerical solvers and computational tools for differential equation systems without built-in physiological structures. A detailed description and comparison of the available software platforms are beyond the scope of this review. Readers interested in a more in-depth overview are referred to the following publications [70–72].

An evaluation of the software platforms used for PBPK model development in the context of antidepressants

revealed a clear predominance of Simcyp® Simulator with nearly half of all identified PBPK models (46%) developed using this platform. Other software tools were used less frequently, including PK-Sim® (often in combination with MoBi®) and GastroPlus®, each accounting for 17% of the identified PBPK models, and R®/RStudio®, which was used in 8% of models. In 12% of the PBPK models, either alternative platforms were applied or no specific PBPK modeling software was reported.

3.5 General Trends in PBPK Model Development Stages

The majority of published PBPK models in psychopharmacology demonstrated at least a moderate degree of methodological characterization and development, with a relatively high proportion of models being classified as “extensive” (29 out of 86) and a comparable proportion categorized as “substantial.” This distribution indicates an overall well-developed methodological landscape of PBPK modeling within this therapeutic area and may reflect increasing awareness and implementation of best-practice recommendations, such as those issued by the US Food and Drug Administration and European Medicines Agency [73, 74], which have progressively shaped expectations regarding model documentation, validation practices, and reporting transparency in recent years. Interestingly, models classified as “extensive” were almost exclusively published from 2018 onwards (with one exception), suggesting a continuous improvement in PBPK modeling practices over time. This trend may be driven by advances in physiological databases, improved mechanistic understanding, increasing regulatory acceptance of PBPK approaches, and ongoing methodological standardization. Notably, only one model was classified as being at the “initial” development stage, while no model reached the highest category (“very extensive”), suggesting that although methodological standards are generally high, fully comprehensive model characterization and validation remain challenging to achieve in practice. The detailed scoring results for each model and category as well as the scoring framework itself are provided in the ESM.

3.6 Antidepressants Used as Perpetrator or Victim Drugs

A large number of studies investigated DDIs in which antidepressants were included as either perpetrator or victim drugs for specific cytochrome P450 (CYP) enzymes. In these studies, PBPK models were not used to characterize the antidepressants themselves, but rather to evaluate interactions involving other drugs and applications of interest.

Consequently, these studies did not meet the predefined eligibility criteria and were excluded during full-text assessment. Nevertheless, the frequent use of antidepressants in PBPK-based DDI investigations requires consideration. In total, 96 of the excluded research articles contained at least one antidepressant as a perpetrator or victim drug within a PBPK modeling framework without the antidepressant being the primary drug of interest or focus. Across these studies, antidepressants were counted 173 times as perpetrator or victim drugs for specific CYP enzymes. This number exceeds the number of articles because individual studies often evaluated multiple antidepressants and single antidepressants were frequently investigated in relation to more than one CYP enzyme.

The most frequently used antidepressants were fluvoxamine ($n = 57$ articles), paroxetine ($n = 27$), fluoxetine ($n = 26$), and bupropion ($n = 19$). In addition, other antidepressants, such as imipramine, sertraline, and amitriptyline, were identified as perpetrator or victim drugs in two to five articles each. When assigned to the CYP enzymes, the most commonly used antidepressant as an inhibitor for CYP3A4 was fluvoxamine ($n = 40$ times), paroxetine as a CYP2D6 inhibitor ($n = 22$ times), fluoxetine as a CYP3A4 inhibitor ($n = 21$ times), and bupropion as a CYP2B6 substrate ($n = 16$ times). Table 3 provides an overview of all antidepressants used as inhibitors or substrates for specific CYP enzymes in at least two articles. Other antidepressants that were used as perpetrator or victim drugs in only one article each were duloxetine (CYP2D6 inhibitor), esketamine (CYP2B6 and CYP3A4 substrate), citalopram and escitalopram (both CYP2C19 substrates), and venlafaxine (CYP2D6 substrate).

3.7 Review Articles

In addition to the primary research articles included in this scoping review, several review articles addressing antidepressants in the context of PBPK modeling were identified during the literature search. As these review articles did not provide additional original data relevant to the objectives of the present review, they were excluded. Relevant primary studies cited within these reviews had already been captured by the systematic literature search and were therefore included elsewhere in this scoping review.

In total, 15 review articles were identified. Only one review focused specifically on antidepressants in the context of PBPK modeling, whereas all other reviews included antidepressants only as part of broader discussions involving multiple drug classes. Notably, a substantial proportion of these review articles addressed drug therapy during pregnancy, frequently highlighting PBPK modeling as a tool for

Table 3 Antidepressants used as perpetrator or victim drugs in at least two or more research articles

	CYP 3A4 inh.	CYP 1A2 inh.	CYP 2C9 inh.	CYP 2C19 inh.	CYP 2D6 inh.	CYP 2B6 substr.	CYP 2C19 substr.	CYP 2D6 substr.	CYP 3A4 substr.
Fluvoxamine	40	14	6	6	5				
Paroxetine	5		1		22			3	
Fluoxetine	21			4	9				
Bupropion					3	16	1		
Imipramine							2	4	1
Sertraline					1			1	
Amitriptyline					1			1	

CYP cytochrome P450, *inh.* inhibitor, *substr.* substrate

Empty cells indicate that the corresponding antidepressant was not used as either a substrate or a perpetrator for this specific CYP enzyme

Table 4 Review articles identified addressing antidepressants in the context of PBPK modeling

Review articles including antidepressants in a PBPK modeling context	
Number of review articles identified	15
Topics covered by review articles	Pregnancy (seven review articles) DDI (three review articles) Pediatric, lactation, drug development, nanoparticles, CNS (one review article each)
Number of review articles with the primary focus on antidepressants	One review article (Zheng et al.: Antidepressants and Antipsychotics in Human Pregnancy: Transfer Across the Placenta and Opportunities for Modeling Studies [75])

CNS central nervous system, DDI drug–drug interaction, PBPK physiologically based pharmacokinetic

dose optimization and exposure assessment. More details are provided in Table 4.

4 Discussion

4.1 Summary of Evidence

Physiologically based pharmacokinetic models have been developed for various antidepressants. However, their availability differs significantly between individual compounds. The number of published PBPK models per antidepressant ranges from zero to ten, which implies a distinct heterogeneity in research intensity. Paroxetine, fluvoxamine, and venlafaxine are the most extensively modeled antidepressants, whereas for compounds, such as mirtazapine or milnacipran, only a single PBPK model or none at all has been identified.

Across all included studies, the predominant objectives of PBPK modeling of antidepressants were to simulate DDIs, pharmacokinetics in pregnant women and drug-(drug-)gene interactions. In contrast, few or no PBPK models addressed other clinically relevant populations, including specific ethnic groups or pediatric patients. The lack of PBPK models for antidepressants in children may be due to general challenges in pediatric model development. As discussed by

Yates et al. [76], reliable pediatric PBPK modeling needs detailed knowledge of the ontogeny of enzymes, transporters, and physiological processes, which is often incomplete or inconsistently reported in children. In addition, limited understanding of disease effects on pharmacokinetics, insufficient data for model validation, and a lack of mechanistic absorption information in children restrict the development of robust models. Practical challenges, including the time-consuming nature of data collection and the need for specialized expertise, further contribute to the slow progress in this area [76]. Together, these factors may help explain why pediatric PBPK models remain scarce, including for antidepressants. Moreover, formal approval of antidepressants for pediatric use remains restricted to a small number of agents, with fluoxetine being one of the few with broad regulatory approval in children and adolescents. This limited approval landscape not only results in frequent off-label prescribing, but also means that clinical data available for model validation is scarce for the majority of antidepressants used in younger patients [77].

A more detailed analysis of model purposes stratified by antidepressant revealed that, for some compounds, PBPK modeling efforts are clustered around a narrow set of research questions. For example, several PBPK models for amitriptyline focus primarily on cardiac pharmacokinetics and toxicity. While such depth within a specific

application area is valuable, it also highlights substantial gaps in the broader clinical context, as other essential scenarios remain largely unexplored even for antidepressants with multiple existing PBPK models.

It is also important to note that certain antidepressants, especially fluvoxamine and paroxetine, are frequently used within PBPK modeling frameworks in the context of DDI studies, acting either as perpetrator or victim drugs. In these cases, the respective antidepressants are often not the primary compounds of interest. Instead, their recurrent use is largely driven by their well-characterized properties as enzyme inhibitors and substrates, which make them suitable methodological test compounds for evaluating interaction mechanisms, rather than by a specific clinical focus on the drugs themselves.

Beyond its methodological dimensions and the clinical aspects already discussed, the direct clinical relevance of PBPK modeling in psychopharmacotherapy warrants further consideration. In this context, PBPK modeling has emerged as a valuable tool to support clinical decision making, particularly in populations underrepresented in conventional clinical trials, such as pregnant women, pediatric and geriatric patients, or individuals with comorbidities. By integrating drug-specific and physiological parameters, PBPK models enable mechanistic prediction of pharmacokinetics in populations where direct clinical evidence is limited or difficult to obtain [78].

Pregnancy represents a particularly relevant application area because gestational physiological changes can substantially alter antidepressant pharmacokinetics and may necessitate dose adjustments. At the same time, untreated maternal depression is associated with adverse maternal and fetal outcomes [79]. Physiologically based pharmacokinetic models may therefore support individualized dose optimization and maternal–fetal exposure assessment during pregnancy [80].

Overall, PBPK modeling may contribute to safer and more personalized antidepressant therapy in vulnerable populations where conventional clinical evidence remains limited. Expanding PBPK modeling efforts to cover a broader range of antidepressants, populations, and clinical scenarios represents an important direction for future research.

4.2 Limitations

Several considerations should be taken into account when interpreting the findings of this scoping review. First, the literature search, screening, and data extraction were primarily conducted by a single reviewer. Although a second reviewer independently screened and verified 10% of all titles and abstracts and extracted at least the same proportion of data, the involvement of additional independent reviewers

screening and extracting the full dataset might have further reduced the risk of selection bias. Nevertheless, the use of a predefined search strategy, the adherence to PRISMA-ScR guidelines, and the discussion of potentially ambiguous cases with a second reviewer support the methodological transparency and reproducibility of the review process.

Second, in line with the objectives of a scoping review, no formal assessment of the methodological quality or predictive performance of the included PBPK models was performed. Consequently, this review does not allow conclusions regarding the performance, comparability, or clinical applicability of individual models. Instead, the focus was placed on mapping the existing body of evidence and identifying areas of concentrated research activity and knowledge gaps, which represent a necessary first step toward future systematic evaluations.

Furthermore, the identified PBPK models varied substantially in their structural design and level of physiological detail. The included models ranged from minimal or semi-physiological approaches, focusing on selected organs or processes, to comprehensive whole-body PBPK models encompassing numerous compartments. Because of this heterogeneity, direct comparisons between models with respect to structural detail or predictive performance were beyond the scope of the present review.

Nevertheless, the authors conducted a systematic evaluation of the model development stage, including the modeling approach, sensitivity, and validation of each PBPK model. Importantly, the model development stage should always be interpreted in the context of the intended model application. For example, in DDI studies or specific clinical scenarios, certain scoring domains may have limited relevance for the predictive performance of the model. In addition, model scores may be constrained by limited availability of experimental data, clinical studies, or parameter inputs. Moreover, heterogeneity in reporting standards across publications may lead to underestimation of model development stage despite adequate underlying model performance.

Additional limitations include the fact that the scoring approach is based on selected criteria and therefore cannot capture all aspects of PBPK model development, physiological plausibility, or predictive performance of the model. Therefore, the proposed framework is intended to identify general trends in PBPK model development stage within psychopharmacology rather than to provide a definitive ranking of individual models.

A further limitation of this review is that relevant PBPK models may not have been captured because of limited accessibility. For example, some models may have been described only in conference abstracts or regulatory submissions without accompanying full-text publications detailing their development and characteristics and were therefore not eligible for inclusion.

Moreover, this review focused on antidepressants approved and commercially available in Germany, reflecting its aim to provide a foundation for clinically relevant research in this context. While this approach improves practical applicability in Germany, it may limit the generalizability of the findings to antidepressants used exclusively in other regions.

Another limitation of this review concerns the clinical interpretability of the included PBPK models. The extent to which modeled pharmacokinetic parameters can inform clinical decisions depends on whether a meaningful exposure–response relationship has been established for the respective antidepressant. While PBPK models describe the pharmacokinetic behavior of a drug in mechanistic terms, their clinical relevance rests on the assumption that plasma concentrations are linked to both efficacy and the occurrence of adverse effects. For tricyclic antidepressants, this relationship is well established [81], as it is for a number of selective serotonin reuptake inhibitors and serotonin and norepinephrine reuptake inhibitors [82–85]. For the latter drug classes, evidence from positron emission tomography imaging studies further supports the notion that antidepressant efficacy is directly related to receptor occupancy in the brain, which in turn depends on the plasma concentration [86]. In contrast, for other antidepressant classes such as monoamine oxidase inhibitors and tetracyclic antidepressants, a consistent concentration–effect relationship has not been clearly demonstrated [87]. Consequently, the ability of PBPK models to inform clinical outcomes varies across compounds and should be interpreted in this context.

5 Conclusions

This scoping review demonstrates that, while PBPK models are available for several antidepressants, substantial gaps remain in both drug coverage and application. For several clinically relevant antidepressants, few or no PBPK models have been published to date. Moreover, even for antidepressants with multiple existing models, the scope of applications is often limited, with models frequently addressing only a narrow range of clinical questions.

Across all identified studies, PBPK models describing specific ethnicities or the use of antidepressants in pediatric patients were scarce. As these populations are frequently underrepresented or excluded from clinical trials because of ethical or practical constraints, the limited availability of PBPK models in these contexts highlights a significant unmet need.

In terms of software platforms used for modeling, Simcyp® Simulator was most frequently used, followed by PK-Sim®/MoBi®, GastroPlus®, and R®/RStudio®. Overall,

the identified distribution of model development stages suggests a generally robust and continuously advancing methodological standard of PBPK modeling in psychopharmacology, while also highlighting remaining opportunities for further model characterization, validation, and reporting transparency.

Future research should aim to address the gaps identified in this review by expanding PBPK modeling efforts to underrepresented antidepressants, populations, and clinical scenarios. In addition, more detailed evaluations of existing models, including assessments of model quality and performance, may further enhance comparability and support the translation of PBPK modeling into clinical and regulatory decision making.

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